Vision Document

AI-Augmented FAIRness Assessment of Research Software: A Hybrid Approach

1.Vision

This thesis develops a prototype system that combines rule-based and AI-based methods to automatically assess the FAIRness of research software repositories on GitHub. Building on existing FAIR4RS indicators and tools such as F-UJI, the system extends current approaches by applying large language models (LLMs) to evaluate qualitative aspects of repository documentation that rule-based tools cannot reliably capture. The novel contribution lies in the systematic analysis of how AI-based methods can complement rule-based FAIRness assessment, including an evaluation of their reliability, limitations, and added value.

2. Introduction / Initial Situation

The increasing importance of open science and reproducibility has led to a growing need for effective management and reuse of digital research outputs, including data, software, and computational workflows. The FAIR principles Findable, Accessible, Interoperable, and Reusable, were introduced to address these challenges by providing guidelines that enhance the discoverability and reuse of research artifacts [1].

While initially developed for research data, the FAIR principles are equally applicable to research software, which plays a central role in modern scientific processes. The FAIR for Research Software (FAIR4RS) principles [3] extend and adapt the original framework to address the specific characteristics of software, such as executability, versioning, and dependency management. Ensuring that software is FAIR is essential for enabling reproducibility, transparency, and collaboration.

However, in practice, many research software repositories hosted on platforms such as GitHub lack essential elements including structured metadata, clear licensing information, persistent identifiers, and comprehensive documentation. Assessing FAIRness at scale therefore remains a complex and largely unsolved challenge.

3. Problem

Current methods for evaluating the FAIRness of research software repositories are limited in their ability to scale and to assess unstructured information. Rule-based approaches, such as those implemented in F-UJI, can effectively detect explicit features such as the presence of a license file or structured metadata, but are insufficient for evaluating qualitative aspects such as documentation clarity, completeness, and usability.

These limitations are addressed in this thesis through a hybrid approach. The rule-based component handles structured, checkable indicators that are already well-defined in the literature. The AI-based component specifically targets the qualitative gaps that rule-based methods cannot fill, namely the assessment of documentation quality, semantic completeness, and reusability-related content in README files and other textual artifacts. By combining both methods and systematically comparing their outputs, this thesis aims to clarify where AI adds value and where its results are uncertain or unreliable.

Furthermore, the lack of automated and scalable solutions makes it difficult to assess large numbers of repositories consistently. A modular prototype focused on GitHub addresses this gap within a scope that is practically feasible for a thesis project.

4. State of the Art

The FAIR principles, introduced by Wilkinson et al. (2016), define a framework for improving the findability, accessibility, interoperability, and reusability of digital research objects [1]. These principles emphasize machine-actionability, enabling automated systems to discover and process research outputs. The FAIR for Research Software (FAIR4RS) principles [3], developed by Chue Hong et al. (2022), adapt this framework specifically to software, accounting for its unique properties such as executability and dependency structures.

Several tools have been developed to automate FAIRness assessment. F-UJI is a prominent example that evaluates repositories against predefined metrics, including metadata availability, persistent identifiers, licensing, and accessibility [4]. The FAIR-IMPACT initiative has further refined these approaches by introducing domain-specific metrics for automated FAIR software assessment [2]. While these tools represent significant progress, they are predominantly rule-based and rely on detecting the presence or absence of explicit repository features. They do not analyze the quality or content of unstructured documentation such as README files, which are critical indicators of reusability and usability.

Research in Mining Software Repositories (MSR) has shown that README files, commit messages, issue trackers, and code comments contain rich information about a project's purpose, usage, and quality [5]. However, this information has not yet been systematically integrated into FAIRness assessment pipelines.

Recent advances in natural language processing (NLP) and large language models (LLMs) offer promising new capabilities for analyzing unstructured text. LLMs such as GPT-4 and similar models have demonstrated strong performance on tasks such as document classification, question answering, and quality assessment [6]. Despite this potential, the integration of LLMs into FAIRness assessment workflows remains underexplored. Existing work has not systematically evaluated whether LLMs can reliably score documentation quality against FAIR4RS criteria, nor has it characterized the types of errors or uncertainties that arise in such assessments. This thesis aims to address these open questions.

5. Requirements and Objectives

The primary objective of this thesis is to design and implement an AI-assisted prototype system for the automated assessment of FAIRness in research software repositories hosted on GitHub, building on and extending existing tools and frameworks. The following requirements are defined:

1. The system must collect and process a representative and diverse dataset of research software repositories from GitHub, following a systematic and reproducible collection process that avoids bias toward specific programming languages or domains.
2. The system must implement measurable FAIR indicators derived from the FAIR4RS principles [3] and mapped to existing frameworks such as those used by F-UJI, rather than defining entirely new metrics from scratch.
3. The system must extend existing rule-based methods with AI-based analysis targeting qualitative FAIR dimensions, specifically documentation quality and reusability, that current tools do not assess. The AI component must be clearly specified in terms of model choice, input data, and scoring approach.
4. The system must generate a FAIRness assessment for each repository, either as a score mapped to existing F-UJI-style metrics or as a clearly defined complementary score, with justification for the design choice.
5. The system must be implemented as a modular Python prototype. Extensibility to other repository platforms such as Zenodo or Codeberg is considered out of scope for this thesis but is noted as a direction for future work.

The thesis addresses the following research questions:

1. To what extent can LLM-based assessment of README documentation improve FAIRness scoring compared to rule-based methods alone?
2. What are the reliability and failure modes of LLM-based FAIRness assessment, and what types of repositories or documentation styles cause errors or produce unreliable results?

6. Proposed Approach

The proposed approach consists of a hybrid system combining rule-based repository analysis with an AI-driven documentation assessment layer. The system will be implemented as a Python prototype, designed to be modular and extensible. A system architecture diagram will be provided in the final thesis to illustrate the data flow and component interactions.

Dataset Collection

A dataset of approximately 100–200 GitHub repositories will be collected using a systematic and reproducible process via the GitHub API. Selection criteria will prioritize diversity across programming languages, research domains, and repository size, rather than limiting to a single language. Repositories will be selected based on relevance to research or scientific computing, as indicated by topics, descriptions, or associated publications. The collection methodology will be documented to ensure reproducibility.

FAIR Indicators

FAIR indicators will be mapped directly to the FAIR4RS principles [3] and aligned with existing frameworks such as F-UJI where possible, rather than defining new metrics from scratch. This ensures comparability with the state of the art. Indicators will cover findability (e.g., metadata, persistent identifiers), accessibility (e.g., repository availability, license presence), interoperability (e.g., standard formats, dependency specifications), and reusability (e.g., documentation quality, citation information).

Rule-Based Component

Rule-based techniques will be used to extract structured information from repositories, replicating and extending what tools like F-UJI already provide. This includes detecting the presence of LICENSE, README, CITATION, and dependency files, as well as checking for persistent identifiers (e.g., DOIs) and structured metadata (e.g., CITATION.cff, codemeta.json). This component serves as the baseline against which the AI-based component is compared.

AI-Based Component (Novel Contribution)

The AI-based component is the core novel contribution of this thesis. It applies an instruction-tuned large language model (LLM) to assess the qualitative content of README files and repository documentation against a structured rubric derived from FAIR4RS reusability and findability indicators. The approach is model-agnostic and allows the use of both commercial models (e.g., Claude, GPT-based models) and open-weight alternatives (e.g., LLaMA-based models), depending on availability and feasibility.

Concretely, the LLM will be prompted to evaluate each repository's documentation on dimensions such as:

- Clarity and completeness of installation and usage instructions
- Presence and quality of contribution guidelines and intended use description
- Description of software dependencies and environment requirements
- Presence of examples or tutorials that support reuse

Each dimension is scored on a scale from 0 to 2, where 0 means not present, 1 means partially present, and 2 means clearly present. This simple scale is used to keep the evaluation consistent. All dimensions are weighted equally. Scores are normalized to allow comparison across repositories.The system produces a FAIRness profile that reports rule-based and AI-based scores separately. The two scores are presented in parallel rather than merged into a single value, ensuring transparency and allowing each component to be interpreted independently. Rule-based methods evaluate structured aspects (e.g., license, metadata), while AI evaluates qualitative aspects (e.g., documentation quality).

The primary focus is on README files as the most consistently available documentation artifact. Where available, additional sources such as wikis or linked documentation pages may also be included. Issues and commit histories are considered out of scope for this prototype but represent a natural extension.

Score Combination

The rule-based and AI-based assessments are reported as part of a combined FAIRness profile rather than merged into a single numerical score. The output highlights which indicators were assessed by rule-based methods and which by AI, along with associated confidence levels.

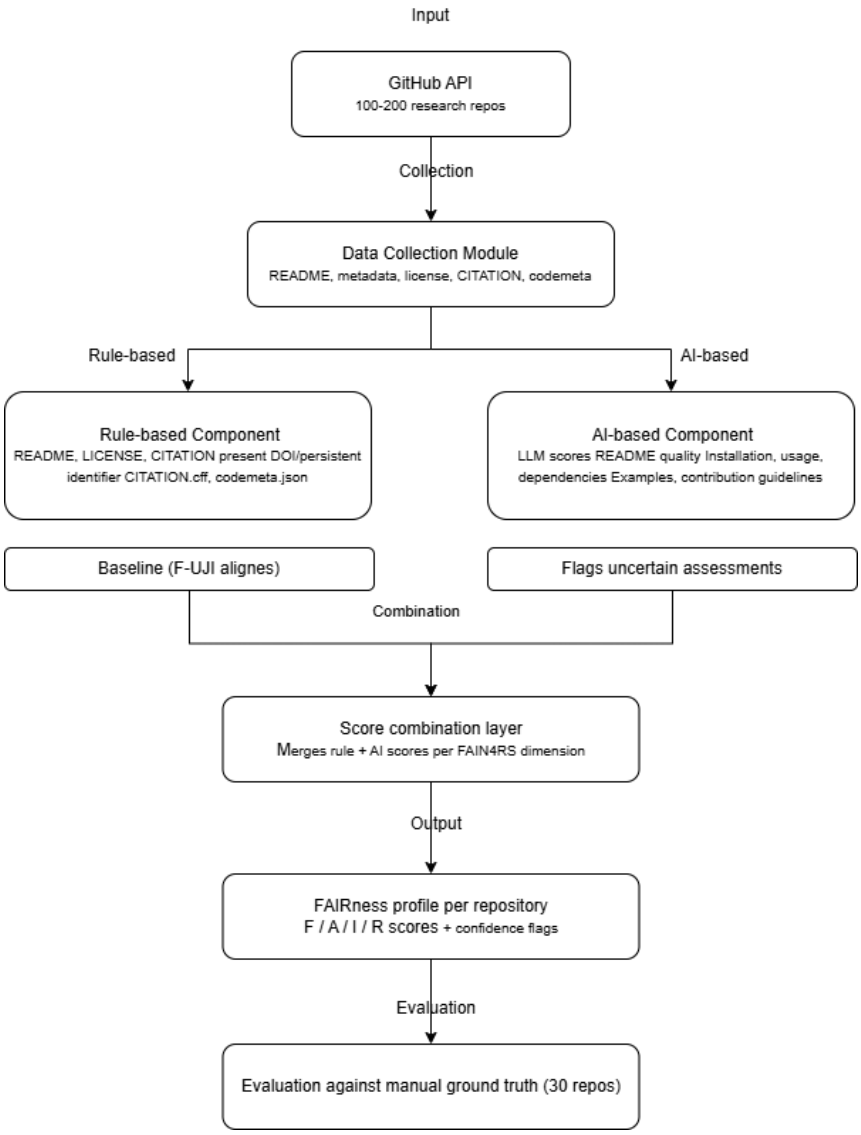


***Figure 1**: Proposed hybrid FAIRness assessment approach*

7. Evaluation Strategy

The evaluation will use both quantitative and qualitative methods to assess the accuracy, reliability, and added value of the proposed system.

Ground Truth Construction

A subset of approximately 30 repositories will be manually annotated by the thesis author using the defined FAIR indicators as a structured rubric. To ensure consistency, annotation will follow explicit guidelines developed prior to the evaluation phase. The use of a single annotator is acknowledged as a limitation and will be discussed explicitly in the thesis.

Quantitative Evaluation

The automated system's outputs will be compared against the ground truth using standard metrics including precision, recall, and F1-score. Results will be reported descriptively, with discussion of patterns and differences between the AI-enhanced system and the rule-based baseline. The AI-enhanced system will also be compared against F-UJI outputs on a shared set of repositories to situate the results within the state of the art.

Qualitative Analysis

A qualitative analysis will identify specific cases where the AI-based component adds value over the rule-based baseline, as well as cases where it fails or produces uncertain results. This systematic error analysis constitutes one of the primary research contributions of the thesis, providing concrete insights into the feasibility and limitations of AI-assisted FAIRness assessment.

8. Novel Contributions

To make the novel contributions of this thesis explicit, the following distinguishes this work from existing rule-based and hybrid approaches:

1. AI-based qualitative assessment: Existing tools such as F-UJI assess only explicit, structured features. This thesis introduces a systematic LLM-based method for scoring documentation quality against FAIR4RS reusability and findability indicators, filling a gap that no current automated tool addresses.
2. Uncertainty quantification: The system incorporates confidence estimation in AI-based assessments, flagging when LLM outputs are uncertain and human review is recommended. This is not present in any existing FAIRness assessment tool.
3. Systematic reliability analysis: The thesis provides a rigorous empirical evaluation of where AI succeeds and where it fails in FAIRness assessment, including comparison with rule-based baselines and contributing reusable insights for future work in this area.

References

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6. T. Brown et al., “Language Models are Few-Shot Learners,” NeurIPS, 2020. (LLM capabilities)